

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT3 — INTERACTIVE DASHBOARD PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Create (BL6)	Assessment:	Assessment Tool 3 – Interactive Dashboard Project
Weightage:	20% of CIA	Total Marks:	25 (Dataset: 7, Interactivity: 9, Effective Visualization: 9)
CO5 Statement:	<i>Create an original, interactive dashboard that selects and prepares an appropriate dataset, implements meaningful software interactivity, and applies effective visualization design principles. (BL6)</i>		
Student Name:	Jothika M	Reg. No.:	192324235
Section / Batch:	Slot C	Date:	27/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This is an individual project. Each student designs, builds, and submits one original interactive dashboard. Total: 25 Marks.
- Electronic devices and software tools are expected and permitted — use any dashboarding tool (e.g., Tableau, Power BI, Plotly Dash, Streamlit, or similar) of your choice.
- Late submissions and academic integrity are governed by the standard course policy; all code/data sources must be cited.
- Submission window: as announced by the instructor (project, not a timed test).
- Choose and clearly justify your own dataset — it must be real or realistic, relevant to a stated purpose, and appropriately cleaned/prepared before use.
- The dashboard must include genuine software interactivity (e.g., filters, zoom/pan, hover tooltips, drill-down, or linked views) — a static image or PDF does not satisfy this requirement.
- Apply effective visualization principles covered in this course (proportional ink, appropriate axis/color scale choice, avoiding overplotting, accessibility) throughout the dashboard.
- Submit both the working dashboard (or a recorded walkthrough, if live hosting is not possible) and a short written justification of your dataset and design choices.

Annexure A – INTERACTIVE DASHBOARD PROJECT

Healthcare Patient Readmission Analysis and KPI Dashboard using Tableau

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Interactive Dashboard Project

CO5 · BT6: Create ·

This rubric is used to assess the CO5-AT3 Interactive Dashboard Project, in which each student designs, builds, and submits an original interactive dashboard, evaluated on dataset quality, software interactivity, and effective visualization design.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset (Selection, Quality & Preparation)	30%	Dataset is highly relevant to a clearly stated purpose, real or realistic, appropriately sized, and thoroughly cleaned/prepared with any limitations disclosed.	Dataset is relevant and reasonably prepared, with only minor gaps in cleaning or justification.	Dataset is loosely relevant or only partially prepared; justification is weak or missing.	Dataset is irrelevant, unprepared, or trivial (e.g., a toy dataset with no real justification).
Interactivity of the Software/Dashboard	35%	Dashboard includes multiple well-implemented, meaningful interactive features (e.g., filters, drill-down, linked views, tooltips) that genuinely help the user explore the data.	Dashboard includes working interactive features that function correctly, with only minor gaps in polish or scope.	Dashboard includes limited or partially working interactivity; some features are superficial or broken.	Dashboard is static or interactivity does not function; the submission does not meet the core project requirement.
Effective Visualization Design	35%	Chart types, color scales, axis choices, and layout are all well-justified and consistently apply the effective-visualization principles covered in this course.	Visualization choices are mostly appropriate and apply course principles, with minor lapses.	Visualization choices show some understanding of course principles but include noticeable design flaws (e.g., misleading axis, poor color scale).	Visualization choices are poor or arbitrary, showing little evidence of applying course principles.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem Statement

Hospitals generate large volumes of patient data on a daily basis — including demographics, chronic conditions, admission types, treatment costs, and follow-up care records. Despite this abundance of data, most hospitals struggle to identify the underlying factors that drive repeated patient readmissions, such as age, chronic disease, length of stay, insurance type, or quality of follow-up care.

Traditional manual analysis using spreadsheets is time-consuming, error-prone, and impractical for large, multi-dimensional healthcare datasets. This makes it difficult for hospital administrators and clinical staff to:

- Identify high-risk patients before they are readmitted
- Detect department- or condition-wise readmission patterns
- Monitor key performance indicators (KPIs) in real time
- Make timely, data-driven decisions to improve patient care and reduce costs

There is a clear need for an **interactive visual analytics solution** that can consolidate raw hospital admission data, compute meaningful KPIs, and present them through an intuitive dashboard — enabling hospital management to monitor readmission performance, identify high-risk patient groups, and support faster, evidence-based clinical and administrative decisions.



Clinical Portal Login

DSA0606 Data Analytics Capstone

STAFF ID / EMAIL

ADMIN-0606

SECURITY PASSWORD

.....

☒ Remember session

[Need assistance?](#)

Sign In to Dashboard

Forgot your password? ID: ADMIN-0606 Password: admin



Patient Readmission Analytics

DSA0606 Capstone Project

Dashboard

Analytics

Patients

Reports

About

[Sign Out](#)

Tableau KPI Dashboard Interoperability Framework

Patient Readmission & KPI Dashboard

Capstone Dashboard Project • DSA0606: Data Handling & Visualization

Reporting Cycle: Last 12 Months

TOTAL PATIENTS



150

100% of total

HIGH-RISK CASES



50

33% of segment

READMISSION RATE



33.3%

0% vs baseline

AVG LENGTH OF STAY



6.5 Days

Target: < 5.0 days

AVG TREATMENT COST



\$21,286

Over budget

RECOVERY/FOLLOW-UP RATE



34.7%

Target: > 75%



Filter Patient cohorts

Segment records for cohort visualization

150 matches

CHRONIC CONDITION

All Conditions

PATIENT GENDER

All Genders

AGE COHORT

All Ages

ADMISSION TYPE

All Admission Types

INSURANCE PLAN

All Plans

READMISSION RISK

All Risk Levels

Reset Filters

Apply Filters



Tableau KPI Dashboard Interoperability Framework

Patient Readmission & KPI Dashboard

Capstone Dashboard Project • DSA0606: Data Handling & Visualization

Reporting Cycle: Last 12 Months

TOTAL PATIENTS



150

100% of total

HIGH-RISK
CASES



50

33% of segment

READMISSION
RATE



33.3%

0% vs baseline

AVG LENGTH OF
STAY



6.5 Days

Target: < 5.0 days

AVG TREATMENT
COST



\$21,286

Over budget

RECOVERY/FOLLOW-
UP RATE



34.7%

Target: > 75%



Filter Patient cohorts

Segment records for cohort visualization

150 matches

CHRONIC CONDITION

All Conditions

PATIENT GENDER

All Genders

AGE COHORT

All Ages

ADMISSION TYPE

All Admission Types

INSURANCE PLAN

All Plans

READMISSION RISK

All Risk Levels

Reset Filters

Apply Filters

HIGH-RISK DEPARTMENT

53.6% risk

COPD Priority

Patients in the **COPD** cohort demonstrate the highest concentration of high readmission risk. Targeted discharge support and early follow-up interventions should be prioritized for this cohort.

Recommended action: Clinical audit of discharge protocols

RISK DRIVER ANALYSIS

4.2x multiplier

Follow-up Care Impact

Data shows that patients labeled with **Poor** post-discharge follow-up care are **4.2 times** more likely to be classified as high risk compared to patients receiving **Good** follow-up care.

Action: Deploy telehealth reminders within 48 hours

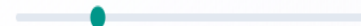
FINANCIAL IMPACT CALCULATOR

Projected Savings

\$175,925

Reduction Target:

15% readmissions



Total Risk Pool Cost:

\$1,172,833

Based on average treatment charges for current high-risk cases

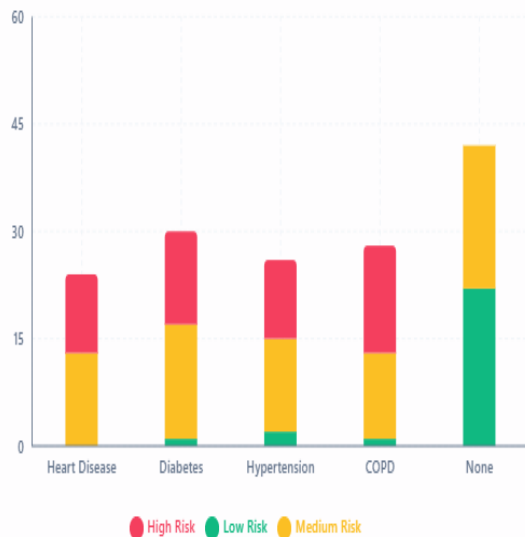


Statistical Visualizations

Cohort breakdown charts mapping clinical attributes

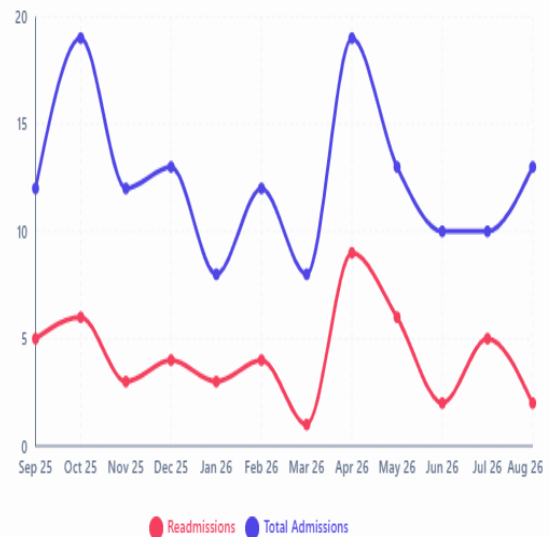
Readmissions by Chronic Condition

Risk categorization sorted by patient medical condition



Monthly Readmission Trend

Total admissions vs high-risk cases longitudinally



Readmission Risk Cohorts

Overview of current segment risk ratios



Age Group vs. Readmission Rate

Matrix view correlating age and chronic risk rates

AGE GROUP	HEART	DIAB	HYPER	COPD
Under 40	50%	30%	17%	50%
40-59	20%	13%	27%	20%
60-74	56%	71%	80%	86%
75+	50%	80%	75%	100%

Insurance vs. Risk Rate

Distribution of readmissions by funding model



High Risk Medium Risk Low Risk



Cohort Registry

Comprehensive index of patients inside selected parameters

Search patient, ID, chronic condition...

Showing 8 of 150 entries

PATIENT ID ↑↓	AGE / GENDER ↑↓	CHRONIC CONDITION	RISK LEVEL ↑↓	LENGTH OF STAY ↑↓	FOLLOW-UP STATUS	ACTION
PAT-0001	73 Yrs Female	Hypertension	High Risk	5 Days	Good Care	Details
PAT-0002	63 Yrs Female	COPD	High Risk	12 Days	Poor Care	Details
PAT-0003	24 Yrs Male	None	Medium Risk	7 Days	Medium Care	Details
PAT-0004	21 Yrs Female	Diabetes	Medium Risk	7 Days	Medium Care	Details
PAT-0005	39 Yrs Male	Heart Disease	Medium Risk	10 Days	Medium Care	Details
PAT-0006	53 Yrs Female	Hypertension	Medium Risk	9 Days	Medium Care	Details
PAT-0007	79 Yrs Female	Diabetes	High Risk	9 Days	Good Care	Details
PAT-0008	72 Yrs Male	None	Medium Risk	2 Days	Medium Care	Details



Tableau KPI Dashboard Interoperability Framework

Patient Readmission & KPI Dashboard

Capstone Dashboard Project • DSA0606: Data Handling & Visualization

Reporting Cycle: Last 12 Months



Filter Patient cohorts

Segment records for cohort visualization

150 matches

CHRONIC CONDITION

PATIENT GENDER

AGE COHORT

ADMISSION TYPE

INSURANCE PLAN

READMISSION RISK

All Conditions

All Genders

All Ages

All Admission Types

All Plans

All Risk Levels

Reset Filters

Apply Filters



Statistical Clinical Correlations

Exposing deep medical dependencies & behavioral coefficients

Automated Statistical Review Active

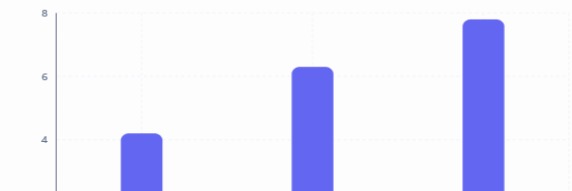
BMI vs. Length of Stay Correlation

Scatter mapping of patient body mass indexes against duration of stay



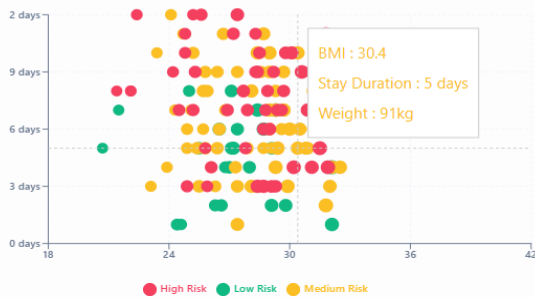
Average Medication Volume by Risk Category

Correlating polypharmacy (medications volume) with readmission rate probability



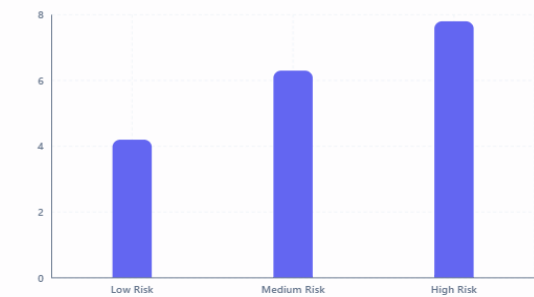
BMI vs. Length of Stay Correlation

Scatter mapping of patient body mass indexes against duration of stay



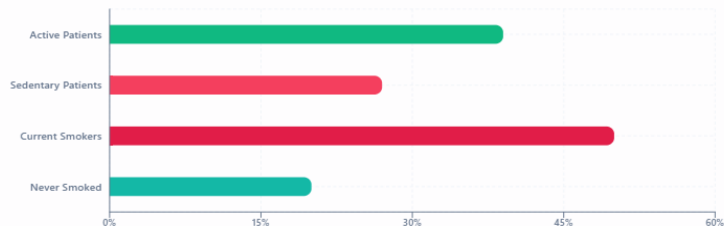
Average Medication Volume by Risk Category

Correlating polypharmacy (medications volume) with readmission rate probability



Behavioral Risk Coefficients

High-risk readmission percentages based on physical activity and smoking status



Clinical Correlation Insight

Based on the vertical coefficient analysis, **sedentary patients** and **current smokers** show a disproportionately high risk rating.

Conversely, patients incorporating **regular physical activity** and **never smokers** average a significantly lower baseline risk rating, indicating behavioral factors are key clinical target points.

Recommended Intervention: Outpatient wellness counseling



Tableau KPI Dashboard Interoperability Framework

Patient Readmission & KPI Dashboard

Capstone Dashboard Project • DSA0606: Data Handling & Visualization

Reporting Cycle: Last 12 Months

Report Configuration

Configure clinical parameters for export

SELECTED CONTENT MODULES

- ☒ Patient Demographics & Vitals
- ☒ Clinical Lab Findings (Hb, Glucose)
- ☒ Chronic Condition Risk Matrix
- ☒ Financial Cost & Savings Model
- ☐ Socio-Behavioral Risk Coefficients

EXPORT FORMAT STYLE

- ☒ PDF Report
- ☐ CSV Data
- ☐ JSON Schema

[Generate Export Package](#)

REPORT READY

[Download PDF](#)

Clinical Cohort Summary Export

Compiled at 8/29/2026, 2:43:31 PM

COHORT SIZE

150

READMISSIONS

50

RISK RATIO

33.3%

AVG STAY

6.5 Days

FINANCIAL LIABILITY MODULE

Gross Cohort Charge Pool: \$3,192,881

Estimated Medicare Penalty Risk: \$95,786 (3% Max Penalties)

INCLUDED CONTENT MODULES:

DEMOGRAPHICS CLINICAL RISKMATRIX FINANCIALS

DSA0606 Capstone Automated Sign-off

[Distribute to Admin Team](#)



Tableau KPI Dashboard Interoperability Framework

Patient Readmission & KPI Dashboard

Capstone Dashboard Project • DSA0606: Data Handling & Visualization

Reporting Cycle: Last 12 Months

DSA0606 CAPSTONE PROJECT

Healthcare Patient Readmission Analysis and KPI Dashboard

A comprehensive data handling and dashboard suite visualizing key risk factors, chronic correlations, and cost overruns to help hospital managers reduce intake rates.



Project Objectives

This capstone project applies data handling, preprocessing, and statistical visualizations to address hospital readmission rates, which are critical metrics of healthcare quality and financial sustainability under the Affordable Care Act's HRRP (Hospital Readmission Reduction Program).



Data & Methodology

Our dataset includes 150+ synthetically generated clinical observations containing demographics (Age, Gender), lab results (Glucose, Hemoglobin, Creatinine), operational variables (Length of Stay, Admission Type), and behavioral metrics. Machine learning heuristics analyze risk indicators based on clinical coefficients.



Tableau Integration

This web dashboard serves as an interactive companion app to our primary Tableau KPI Dashboard. While Tableau handles complex visual cross-filtering and multi-dimensional aggregations, this React dashboard provides real-time client-side cohort calculations and clinical simulation capabilities.



Course Information & Guidelines

Course Name:

DSA0606 – Data Handling and Visualization

Focus Area:

Predictive Analytics & Tableau Interoperability

Deliverables:

Tableau Packaged Workbook, Data Pipeline Script, UI Mockup